

$$4-1) \quad T_1 = 540 \text{ K} \quad T_2 = 300 \text{ K} \\ Q_1 = 1000 \text{ J} \quad W_{\text{net}} = 450 \text{ J}$$

$$W = Q_1 - Q_2 \quad Q_2 = Q_1 - W = 550 \text{ J} \quad \eta_{t,c} = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{540} = 0.4444 \\ \eta_t = \frac{W}{Q_1} = 0.45$$

$$\eta_t > \eta_{t,c} \Rightarrow \text{设计不合理}$$

$$4-3) \quad \text{高温 } 150^\circ\text{C}, \text{ 低温 } 10^\circ\text{C}$$

$$(1) \text{ 热机效率 } \eta_t = \frac{W_{\text{net}}}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} = 1 - \frac{273+10}{273+150} = 0.331$$

$$(2) \text{ 热机输出功 } W_{\text{net}} = 2.7 \text{ kJ} \quad Q_1 - Q_2 = 2.7 \text{ kJ} \\ \frac{2700}{Q_1} = 0.331 \quad Q_1 = 8157.0997 \text{ J} \quad Q_2 = Q_1 - 2700 \\ = 5457.0997 \text{ J}$$

$$(3) \text{ 供热系数 } \varepsilon' = \frac{T_1}{T_1 - T_2} = \frac{273+150}{273+150 - 273+10} = \frac{423}{140} = 3.0214$$

$$4-7) \quad m \text{ 相同}, c \text{ 相同为常数} \quad T_A - \text{高温热源} \quad T_B - \text{低温热源}$$

$$(1) \text{ 平衡时}, \Delta S_{\text{iso}} = \Delta S_{fA} + \Delta S_{fB} + \Delta S_g \quad \text{可逆过程} \Rightarrow \Delta S_g = 0 \\ 0 = mc \ln \frac{T_m}{T_A} + mc \ln \frac{T_m}{T_B}$$

$$0 = \ln T_m - \ln T_A + \ln T_m - \ln T_B \\ \ln T_A T_B = \ln T_m^2 \\ T_m = \sqrt{T_A T_B}$$

$$(2) \quad W = Q_A - Q_B = mc [(T_A - T_m) - (T_m - T_B)] \\ = mc (T_A + T_B - 2T_m)$$

$$(3) \quad T_m = \frac{T_A + T_B}{2}$$

$$\begin{aligned} \Delta S_{\text{total}} &= \Delta S_{f_A} + \Delta S_{f_B} = mc \ln \frac{T_m}{T_A} + mc \ln \frac{T_m}{T_B} \\ &= mc \left(\ln \frac{T_m}{T_A} + \ln \frac{T_m}{T_B} \right) \\ &= mc \ln \left[\frac{(T_m)^2}{T_A T_B} \right] \end{aligned}$$

$$4-10) \quad (1) \quad Q = mc_p \Delta T = mc_p (T_1 - T_2) = 6(1400)(527 - 37) = 4116 \text{ kJ}$$

(2) $Q_{2, \min}$, 过程可逆, 取烟气工质和低温热源为孤立系统。
利用孤立系统的可逆过程熵不变原理可得:

$$\Delta S_{\text{total}} = mc_p \ln \frac{T_2}{T_1} + \frac{Q_2}{T_0} + 0 = 0$$

$$\begin{aligned} Q_2 &= -T_0 mc_p \ln \frac{273+37}{527+273} = -(27+273) \times 6 \times 1400 \times \ln \frac{310}{800} \\ &= 2389.0594 \text{ kJ} \end{aligned}$$

$$4-13) \quad \Delta S = \Delta S_1 + \Delta S_2$$

$$= mc_p \ln \frac{T_m}{T_1} + mc_p \ln \frac{T_m}{T_2}$$

$$T_m = \frac{T_1 + T_2}{2}$$

$$= mc_p \ln \left(\frac{T_m^2}{T_1 T_2} \right)$$

$$= mc_p \ln \left(\frac{(T_1 + T_2)^2}{4 T_1 T_2} \right)$$

$$= mc_p \ln \left(\frac{(T_1 + T_2)^2}{4 T_1 T_2} \right)^{\frac{1}{2} \cdot 2}$$

$$= 2mc_p \ln \left(\frac{T_1 + T_2}{2 \sqrt{T_1 T_2}} \right) \quad \square$$

$$4-18) \quad C_{p,m} - C_{v,m} = R$$

$$C_{v,m} = C_{p,m} - R = 29 - 8.314 = 20.686 \text{ J/mol} \cdot \text{K}$$

$$k = \frac{C_{p,m}}{C_{v,m}} = \frac{29}{20.686} = 1.4019$$

$$1 < \gamma = 1.40 < k = 1.402$$

$\therefore$ 吸热过程